A Framework for Digital Forensics Using Blockchain to Secure Digital Data

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***Abstract*— Digital forensics (DF) requires evidence integrity and provenance across boundaries of jurisdiction, and blockchain technology is ideal for ensuring that. As part of this paper, we discussed a digital forensic framework designed to help prevent duplication of data and secure digital data. In order to accomplish such forensic capabilities, we provide a block-based forensics framework. Using it, examinations are validated, irreversible, traceable, robust, and demonstrate high levels of confidence among examiners and evidence entities.**

Keywords*— *Blockchain, Digital forensics, Digital Evidence

I. INTRODUCTION

IoT (Internet of Things), decentralized blockchains, 5G (5th generation) connectivity, and social media have become integral parts of contemporary life. Modern technologies make life simpler, faster, more engaging, producing innovative tools, gadgets and resources that make the most critical information readily accessible. Technology has made it easier for criminals to operate in an environment where many devices are connected to the Internet. These new technologies make cybercrime more difficult to identify and prosecute than conventional crimes, according to studies. In forensic investigations linking persons to illegal conduct, digital evidence will play an increasingly significant role. In an IoT context, maintaining auditability, traceability, and evidence integrity is important.

Cyber-physical systems present additional difficulties to current DF, including data inaccessibility from different sources, data provenances at multiple locations, evidence transparency, traceability, and large-scale data analysis. In the IoT environment, cyber-physical systems (CPS) can provide a wide range of control for complex industrial systems through heterogeneous architectures of connected sensors and devices. Medical records and monitoring patients' health are among the tasks that IoT devices perform that are security-critical. Therefore, attackers are very interested in IoT-based systems. In the event that these systems are successfully attacked, users' lives could be at risk. If a patient's IoT-enabled pacemaker is compromised, it might be life-threatening. Vehicles equipped with intelligent sensors may also be vulnerable to attacks. Cybercriminals may even hack into smart home products to spy on people, such as security cameras. It is necessary to define a framework for distributed systems (DF) for new challenges posed by distributed systems environments, to guarantee digital evidence reliability, availability, and durability in an increasingly complex environment, and to comply with new privacy laws, such as the General Data Protection Regulation (GDPR). New DF research needs to address these procedural, social, and legal difficulties. Security of digital data is a top priority when it comes to cybercrime. Forensics plays an important role in this area.

A blockchain is a decentralized database in which entries are stored by peers via a peer-to-peer network. Data is stored as timestamped blocks linked in a chain, resulting in an immutable audit trail that can be viewed by anyone and can be verified by a consensus proof of trust. Each transaction is inextricably linked to the previous one in this decentralized, unchangeable ledger. Transaction records and decentralized consensus on the authenticity of the transaction record can safeguard operations in untrusted decentralized contexts, which is the central function of blockchain technology. A transaction may also include operational code, allowing software services to be shared between untrusted users. It has become apparent that blockchain technologies have grown in interest since they were initially used in cryptocurrencies, as industry and academia evaluate the applications and functioning of the underlying technology. Several emerging application demands have been identified as potential blockchain applications, including Internet of Things (IoT), Big Data, Cloud and Edge Computing, Identity Management, and others. As part of this effort, the industry is evaluating the usefulness of blockchain for different commercial uses, and various frameworks are being developed to support these future applications.

A distributed architecture using cryptographically signed transactions operates on a block-by-block basis. A cryptographic method is associated with each block. Every point of failure should be examined and verified to confirm the legitimacy of the transaction. The peer-to-peer model has many characteristics of the P2P model. In addition to the hash value of the current and previous blocks, the time stamp, and some other information, such as the index, the blockchain structure contains a number of other elements. Using the first block hash value, the previous initial hash value is null. As the previous hash for the second block, this hash is added to the second block. The chain is built and connected.

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A. *Chain of custody (CoC)*

In Digital Forensics Chain of Custody (CoC), evidence is tracked and documented through a detailed history. As a piece of digital forensic evidence passes through different echelons of hierarchy, including the first responders and the higher authorities overseeing cybercrime investigations, the Chain of Custody preserves all the minute details. In addition to tracking how evidence was acquired, analyzed, and maintained for production, CoC keeps track of when, where, and who came into contact with it. The forensic CoC may, however, be compromised if proper documentation is not properly maintained. This will render the CoC inadmissible in court as evidence in criminal cases. For this reason, the digital forensic Certificate of Completion must maintain its integrity in order to be accepted in court.

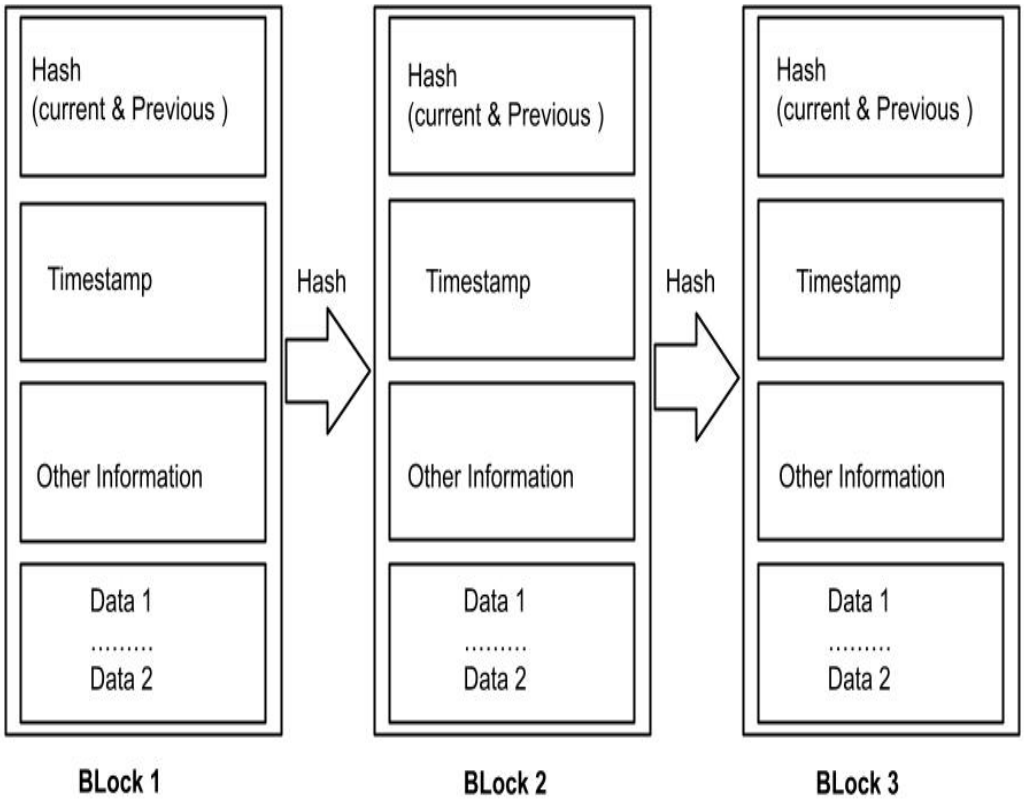


Fig. 1. Structure of blockchain

II. LITERATURE REVIEW

According to Li et al. [1], they created a distributed consensus framework to optimize the LLRs and choose the global optimum. The consensus issue seems to be one of the most important factors affecting the performance of consensus-based applications. Using public ledgers for forensic investigation, Hasan et al. [2] propose an IoT-based infrastructure for digital forensic investigation. This ledger tracks interactions between devices, devices and users, and devices and the cloud in a way similar to that of Bitcoin. By utilizing FIF-IoT, evidence made publicly available may remain secret, anonymous, and nonrepudiable. An FIF-IoT interface may also be used during the investigation of a criminal occurrence to gather evidence and validate its integrity. In this case, no analysis of the evidence is conducted. Hossain [3] and colleagues proposed a digital forensic architecture for IoT systems that includes procedures for collecting and storing reliable evidence from dispersed infrastructure. The evidence analysis has not been completed.

According to Wu et al., [4] the blockchain-based e-auction system they presented was built to reduce collusion. Privacy preservation is a disadvantage of this method. The cloud platforms provided by Ritzdorf et al. [5] are utilized as shared repositories, but they do not support the concept of shared ownership. They developed a unique idea of shared ownership in this research, which we defined using a formal access control model.

In analyzing privacy breaches induced by malware in cloud computing settings, Zou et al. [6] recommended a tool with multi-granularity privacy leakage forensics capabilities.

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Dynamic taint analysis and RAM mirroring technologies will assist with the forensics evaluation. Here, we use basic characteristics to achieve a clearer and more accurate leakage of private information. Sudhakar et al. [7] In this research, they show how to effectively update many incremental data sets while meeting data owners' privacy needs. Over incremental and dispersed data sets, an index-based quasi-identifier technique ensures privacy protection and high data usefulness.

Using the blockchain ledger in conjunction with distributed queueing systems, Pop et al. [8] proposed a scalable second-tier solution to reduce the need to register energy transactions frequently on the blockchain and maintain the tamper evidence offered by blockchain technology. Using a blockchain, Shekhtman and Waisbard [9] created a tamper-resistant log system. They use the blockchain's immutable write action, distributed storage, and an extra encryption layer to create a secure log system. Using artificial communities, computational experiments, and parallel execution, Wang et al. [10] created a parallel crime scene analysis system (ACP). Zhang et al. [11] propose a process provenance model that takes advantage of blockchains and cryptographic group signatures to offer evidence of the existence of process records while protecting their privacy.

A trade-off study model of the CSOC was developed by Shah et al. [12] to evaluate and quantify the interactions among the qualities described above and to provide basic principles for the establishment and operation of an effective CSOC based on the information gathered from that study. The features of the CSOC mentioned previously are optimized by using a constraint-optimization trade-off study model, which is then evaluated through multiple simulations of alert arrival and service operations at the CSOC. Liu and Seo[13] used standard, constant-time execution to defend against timing and basic side-channel attacks to introduce NUMS256, NUMS379, and NUMS384 curves on 8-bit AVR and 32-bit ARM11 microcontrollers; the implementations achieved speed records and advanced the state-of-the-art curve-based (without endomorphism) scalar multiplication algorithm. Researchers Yong and colleagues explored the privacy and security challenges associated with IoT. An open source framework for integrating blockchain with the IoT offers high assurance for IoT data, as well as a variety of functionality and scalability, such as decentralized payment and authentication. They also demonstrate how blockchain could help IoT by proposing several blockchain-based solutions to these challenges. Providing forensic information and anti-forensic tactics, Joseph and Norman [15] offered users and researchers insight into forensics' many facets, and they evaluated where the field stands today. They introduced FoBI, an IoT-based fog-based forensic framework.

The work by Al-Masri et al. [16] addressed the challenges of digital IoT forensics. FoBI's general design, application cases, and implementation specifics have been discussed. In addition to providing insights into improving digital forensics procedures, we're also using FoBI techniques to improve IoT device forensic procedures. Amato et al. [17] (MULTI-FORESEE) describe in this European COST Action the applications of face recognition and verification to forensic science. This study compares and contrasts different approaches to facial recognition and verification, from traditional facial landmark identification to state-of-the-art Convolutional Neural Networks (CNN) off-the-shelf and CNNs specifically educated for the task at hand.

III. METHODOLOGY

Before, the blockchain cloud storage system broke the user's data into minor parts. Then they add another layer of protection and disperse it around the network. Blockchain technologies like hashing, private/public key encryption, and transaction data make this feasible (ledgers). Because many small companies engage in cloud storage by contributing their processing power and space to store data, blockchain is the newest and perhaps cheapest option to access cloud storage. Blockchain is a growing series of documents called blocks that are cryptographically connected. Each block includes the preceding block's crypto, a timestamp, and transaction data. Many IoT nodes capture and handle private data, making them a goldmine of information for hostile actors. In order to successfully deploy IoT networks, security, particularly the capacity to identify compromised nodes, and gather and retain evidence of an attack or malicious activity, become a priority.

When a crime is committed, physical and digital evidence play the most important roles in the investigation. Competent computer investigators will be required as the field of digital investigation grows, especially as they are involved in capturing the crime scene, calling-in data records, examining gathered records, retrieving data, and participating in the forensic process. The nature of digital evidence makes it difficult for forensics teams to work with it. Evidence that has been altered and updated with new data, or that has been removed of critical evidence, easily replicated or reproduced. In certain instances, there may be a time-sensitive nature to the case or pre-arrest conditions that make evidence access, secure evidence storage, and digital evidence difficult. When digital evidence is delivered in court instead of physical evidence, a problem arises. There is a serious threat to the reliability and security of digital witnesses in the legal system. Digital forensics models and techniques are also diverse, posing another problem.

Collecting digital evidence from the crime scene is the framework's first step. Images, logs, text, audio, and other data types may be used. Following the acquisition of digital data, the next step is to upload it to a database for safe storage and access to the evidence. Once the data has been saved, the data's hash is calculated. To find the duplicate, the hash is compared to the existing hash. The hash value is used to create the block.

Following are six steps to demonstrate the chain of custody in digital forensics using blockchain technology.

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Additionally, Fig. 2 and 3 illustrate flow diagram of block chain and workflow used in this work respectively.

1. Evidence is gathered from the crime scene or the investigation location in text or system logs that include the time the evidence was obtained to create a timeline.
2. The obtained data is uploaded to a database, which aids in storing case information. The produced data is taken and utilized in the blockchain for hashing.
3. The retrieved data is converted to a string and hashed using the hash algorithm.
4. The generated hash is matched with the existing hash directory to find the duplicated data.
5. The timestamp is added to the block. The timestamp assists in determining when the proof was submitted to the blockchain. It will alter if it is tampered with, causing the chain to break. If the chain is not broken, the block is in good condition.
6. PoW is a way of determining if evidence has been tampered with since the blocks' connections would have been lost at a certain point. This is accomplished by rehashing the previous blocks to match the current data.

The initial data collection is important in cybercrime as it plays an important role in solving it. The strong digital evidence led to the best outcome. Storing this evidence in the blockchain as a block help to inherit the integrity of digital evidence. The first stage of the system is collecting digital evidence from the crime scene. The collected digital evidence is hashed; we use SHA 256/512 to hash the data. This hash is matched in the hash directory to find the data duplication and then added to the blockchain.

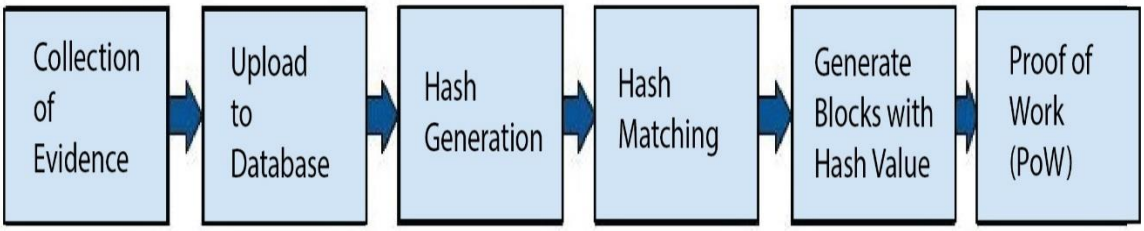


Fig. 2. Flow diagram

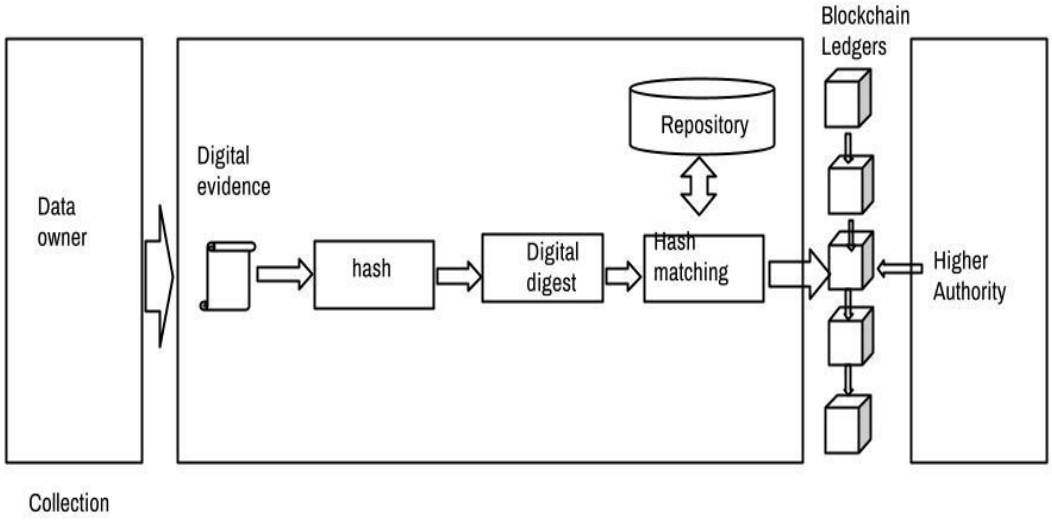


Fig. 3. Workflow of the proposed system

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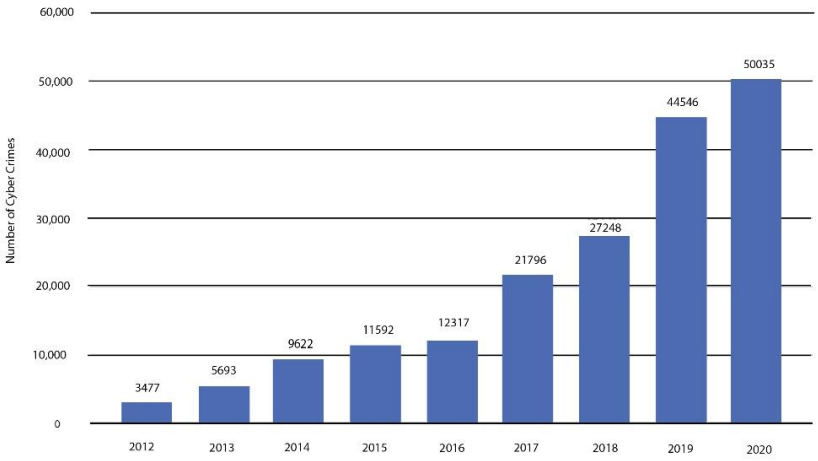


Fig. 4. Cybercrimes reported across India from 2012 to 2020

IV. Result Analysis and Discussion

We propose a method of maintaining and certifying the integrity of digital evidence using distributed ledger technology, thus enhancing the trust in the authorities handling it. The blockchain maintains the data securely. To enhance forensic applications, blockchain-based DF chains of custody can ensure the integrity, transparency, authenticity, security, and auditability of digital evidence to accomplish the intended objective. This chain can assist in collecting, preserving, and validating evidence. Law enforcement may also benefit from blockchain technology in tracking, monitoring, and apprehending cyber offenders more effectively.

When new technologies emerge, cybercrime hits a new high point every year. If we look at cybercrime reporting in India over the last decade, we can observe that it has increased many times, as shown in graphs plotted on the statistics of cybercrimes in India. There is a sharp rise in cybercrime in this country. The data shows that from 2012 till 2020 (refer Fig 4), it shows an exponential rise. This data proves the importance of digital forensics and digital evidence security, an integral part of the prosecution should be this data.

V. Conclusion

Blockchain technology has a wide range of applications in today's environment. In the subject of cybercrime, it is also used and explored. Blockchain architecture is a viable answer to today's security concerns in IoT device chains, data transport, and storage in the digital forensics field. Blockchain can protect data during transmission and storage by using encryption available to all parties and is not susceptible to manipulation.

On its own, technology will never be sufficient to combat cybercrime. For recognizing and detecting illegal activity, human specialists and experience are required. Here, blockchain technology will aid in the safe management of data and provide a platform for professionals to work efficiently, allowing us to overcome cyber-crime difficulties. As a result, BlockChain might play a role in a long-term solution.

According to the report, future academics should focus on the feasibility of a single BlockChain for developing security rather than the existing scenario.

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